
Software Requirements Specification (SRS)

For: CRAMS - Course Registration and Advising Management System

1. Introduction

1.1 Purpose

The purpose of this SRS is to define the requirements for the CRAMS, an integrated web-based platform designed to streamline course registration and academic advising processes for universities. The system will serve students, advisors, and administrators by automating key workflows, providing real-time data, and improving overall efficiency.

1.2 Scope

CRAMS will eliminate manual bottlenecks and fragmented communication by offering:

- Unified registration and scheduling
- Real-time seat and conflict tracking

- Automated advisor review and feedback processes
- Role-based dashboards for students, advisors, and admins
- Integrated notifications and reporting

1.3 Definitions, Acronyms, Abbreviations

- **CRAMS:** Course Registration and Advising Management System
 - **UI:** User Interface
 - **Admin:** Administrator
 - **CGPA:** Cumulative Grade Point Average
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2. Overall Description

2.1 Product Perspective

CRAMS is a new, modern university registration and advising solution, replacing ad-hoc legacy processes. It integrates frontend (React.js, Tailwind CSS), backend (Node.js/Express), database (MongoDB), and secure authentication (JWT).

2.2 Product Functions

- Role-based authentication (Student, Advisor, Admin)

- Course selection with real-time seat and conflict checking
- Advisor approval/rejection with feedback features
- Admin tools for managing courses, sections, seats, user accounts, and system settings
- Central dashboards summarizing progress, deadlines, and quick actions for every role

2.3 User Classes and Characteristics

- **Students:** Register for courses, plan schedules, view statuses, and receive notifications
 - **Advisors:** Review and approve/reject course requests, manage advisees, provide feedback
 - **Admins:** Control course offerings, sections, users, policies, and monitor all system activity
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3. Functional Requirements

3.1 User Authentication and Role Management

- System must authenticate users securely and assign appropriate roles (student, advisor, admin)

- Support secure password management and password reset

3.2 Course Registration and Selection

- Allow students to browse/select courses with updates on seats and prerequisites
- Prevent students from submitting schedules with time/prerequisite conflicts
- Display real-time availability counts, waitlisting if full

3.3 Advisor Review Process

- Advisors must be able to view and approve or reject student registrations
- Feedback must be visible to the student in their status dashboard
- Notifications sent for approvals, rejections, and pending actions

3.4 Section and Seat Management

- Admins set capacities for sections; system tracks enrolled and available counts

- Waitlist is managed and automatically notifies students of seat openings

3.5 Academic Planning Tools

- Students have access to dashboards showing completed/pending credits, registration progress, and upcoming deadlines
- Advisors see aggregated advisee data with status indicators and averages

3.6 Notification System

- System supports real-time notifications and reminders (email, SMS) for all key transactions (approvals, deadlines, changes)

3.7 Additional Admin Functions

- Manage users (add/edit/delete), assign advisors, and configure system-wide settings (semester, policies, maintenance mode)

4. Non-Functional Requirements

4.1 Performance

- Web application should support at least 1,000 concurrent users
- All key actions (login, search, update) must complete in under 2 seconds

4.2 Security

- Role-based access control for all functionality
- Passwords stored encrypted, session tokens not exposed

4.3 Reliability & Availability

- System must be available 99.5% of the time, with full data backup and disaster recovery compliance

4.4 Usability

- Interfaces should be intuitive, consistent, and accessible to users with varying levels of technical expertise
- Mobile-responsive design preferred

4.5 Maintainability

- Modular codebase with documented APIs and deployment scripts
- Admin UI for changing most configurations without code changes

5. System Features (UI Mapping Example)

Feature	UI Reference	Description
Login/Access	Login screen, role tabs	Secure login, select user role
Student Dashboard	"Welcome back, [name]"— Registered Courses, Progress Bar, Deadlines	Overview, credit counts, quick links, notifications
My Schedule	Weekly grid of courses, rooms, and details	Visual timetable and details
Course Selection	List with search, real-time seat and conflict indicators	Register, handle prerequisites/warnings
Registration Status	Approved, pending, rejected tabs with feedback history	Track approval process & advisor comments

Pending Reviews (Advisor)	Pending courses requiring advisor action, feedback on issues	Bulk approve/reject, see student stats
Admin Management	Users, sections, courses, system settings dashboards	Add/edit courses/sections, assign users, view audit trail

6. Future Enhancements

- AI-powered course recommendations
- Degree audit and graduation tracking
- 24/7 chatbot for support
- Predictive analytics using machine learning

7.Diagram

